MASTER OF COMPUTER SCIENCE
BINUS GRADUATE PROGRAM
BINA NUSANTARA UNIVERSITY
JAKARTA

ASSESSMENT FORM

Course: COMP8044041 - Deep Learning and Its Applications

Method of Assessment: Project

Semester/Academic Year : 1/2025-2026

Name of Lecturer : **I Gede Putra Kusuma Negara, B.Eng., PhD**

Date : **……**

Class : **……**

Topic : **Deep Learning for Computer Vision**

Group Members :	1_____ 2_____ 3_____ 4_____ 5_____ 6_____ 7_____ 8_____

Student Outcomes:

(SO 5) Mampu mengembangkan ilmu pengetahuan dan Teknologi Informasi dan Komunikasi dengan menggunakan metode kecerdasan buatan untuk menghasilkan produk-produk inovatif yang dapat diaplikasikan dalam berbagai bidang.
Able to develop science and Information and Communication Technology using artificial intelligence methods to produce innovative products that can be applied in various fields.

Learning Objectives:

(L.Obj 5.3) Mampu mengusulkan penelitian inovatif.
Able to propose innovative research.

Learning Outcomes:

- LO4: Analyze the architectures and performances of deep learning models (L.Obj 5.2 and L.Obj 5.3).
- LO5: Evaluate the advancements and challenges in deep learning research (L.Obj 5.3).
- LO6: Design new approaches that can improve the deep learning performances (L.Obj 5.3).

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (90 - 100)	Very Good (85-89)	Good (80-84)	Average (75-79)	Poor (0-74)	Score	(Score x Weight)
1	LO4 – L.Obj 5.2 – SO 5	Combination and development of research	10%	Able to demonstrates comprehensive ability to evaluate deep learning model performance using a full suite of industry-standard metrics and validation techniques.	Mostly able to evaluate the performance of deep learning models, with minor gaps in applying industry best practices or interpreting results.	Can evaluate basic performance metrics of deep learning models, but lacks depth in understanding or applying industry best practices.	Attempts to evaluate model performance, but relies on incomplete or inappropriate methods and shows limited understanding of key evaluation concepts.	Unable to appropriately evaluate deep learning model performance, misinterprets key metrics or ignores standard evaluation protocols, leading to inaccurate or misleading conclusions.		
2	LO4 – L.Obj 5.3 – SO 5	*Able to propose innovative research.*	30%	Demonstrates the ability to implement and adapt multiple (more than two) state-of-the-art deep learning architectures.	Capable of implementing two established deep learning architectures using standard libraries.	Able to implement at least one well-known deep learning model with guidance or reference.	Can replicate simple deep learning architectures from tutorials, but shows limited understanding of underlying components.	Unable to implement deep learning architectures independently; lacks foundational understanding of model structure, training workflows, or appropriate use of frameworks.		
3	LO5 – L.Obj 5.3 – SO 5	*Able to propose innovative research.*	30%	Demonstrates a full understanding of both recent	Identifies key advancements and challenges in deep learning	Can recognize a few major trends or challenges in deep learning but	Demonstrates a superficial or fragmented understanding of	Unable to identify or evaluate significant advancements or		

No	Related LO-LOBJ-SO	Assessment criteria	Weight	Excellent (90 - 100)	Very Good (85-89)	Good (80-84)	Average (75-79)	Poor (0-74)	Score	(Score x Weight)
				advancements and ongoing challenges in deep learning research.	research and can explain their significance with some reference to current developments	provides limited analysis.	deep learning research.	challenges in deep learning research or shows fundamental misconceptions.		
4	LO6 – L.Obj 5.3 – SO 5	*Able to propose innovative research.*	30%	Consistently designs innovative and well-justified approaches that significantly enhance deep learning performance.	Able to design new or adapted approaches that lead to moderate improvements in deep learning performance.	Designs approaches with minimal novelty or impact and lacks clear justification.	Struggles to design meaningful improvements and relies on trial-and-error changes without theoretical basis.	Unable to design any viable approach to improve deep learning performance.		
	Total Score: ∑(Score x Weight)									

Remarks:

ASSESSMENT METHOD

Instructions:

1. Project is worked on individually.
2. Select a **case or topic in the field of COMPUTER VISION** that will be addressed through the development of a deep learning model. Describe the case or topic and why it is interesting to address. Then, identify the dataset used to conduct the experiment. Explain the composition of the dataset and provide examples of the data.

Implement **two different deep learning models** to solve the selected case or topic in the field of computer vision. The two models can be different or an improvement on the first. Conduct experiments to evaluate the performance of both models. Report the two implemented deep learning models and their experimental results. **Compare the performance of the two evaluated models** and provide an analysis.

Project Output:

Project report submission in the form of a final report document in .docx or .pdf format and implementation source code in the Python programming language.

Note for Lecturers:

1. Lecturers can provide direction regarding the format and structure of project reports to students.
2. Lecturers **are not permitted** to require students to make project reports in scientific article format for submission to journals or conferences.

Assessment Activity Weight Score:

Assessment Activity	Weight
PROJECT I (AoL)	**50%**
PROJECT II (AoL)	**50%**